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Heart failure

Definitions

- ❖ Heart failure (HF) is a progressive clinical syndrome that can result from any changes in **cardiac structure or function that** impair the ability of the *ventricle to fill* with or *eject blood*.
- ❖ Occurs When the ability of the heart to effectively pump blood to the rest of the **body is compromised**, this will result in a **mismatch** between **supply** and **demand** of the metabolizing tissues during ordinary activity.
- ❖ Congestive heart failure can occur as a result
 - Impaired **cardiac contractility**, which is known as systolic congestive heart failure
 - myocardial stiffness or an inability of the **heart to relax** in the absence of *reduced contractility*,

Ethology of Heart Failure

❖ HF can result from any disorder that affects the ability of the heart to contract (systolic function) and/or relax (**diastolic dysfunction**).

❖ Table: Common causes of HF

TABLE 4-1 Causes of Heart Failure

Systolic dysfunction (decreased contractility)

- Reduction in muscle mass (e.g., myocardial infarction)
- Dilated cardiomyopathies
- Ventricular hypertrophy
 - Pressure overload (e.g., systemic or pulmonary hypertension, aortic or pulmonic valve stenosis)
 - Volume overload (e.g., valvular regurgitation, shunts, high-output states)

Diastolic dysfunction (restriction in ventricular filling)

- Increased ventricular stiffness
 - Ventricular hypertrophy (e.g., hypertrophic cardiomyopathy, other examples above)
 - Infiltrative myocardial diseases (e.g., amyloidosis, sarcoidosis, endomyocardial fibrosis)
 - Myocardial ischemia and infarction
- Mitral or tricuspid valve stenosis
- Pericardial disease (e.g., pericarditis, pericardial tamponade)

Stages of Heart Failure

AHA-HF classification

Stage A

- People who have underlying risk factors for developing heart failure
- Do not have any structural abnormalities of the heart.
- Patients with hypertension, dyslipidemia, diabetes mellitus, obesity, smoking, metabolic syndrome, and exposure to chemotherapy that can be cardio toxic.

Stage B

- Includes people who have structural abnormalities of the heart
- Do Not exhibit signs or symptoms of heart failure.
- These include patients with underlying left ventricular hypertrophy, myocardial infarction, systolic dysfunction, and valvular heart disease.
- These are detected by electrocardiography (EKG) or echocardiography (ECG).

Stages of Heart Failure...

Stage C

- patients who have signs and symptoms of heart failure.
- patients who are **currently asymptomatic** but may have had symptoms previously and are currently on medications such as β -blockers and angiotensin-converting enzyme (ACE) inhibitors.

Stage D

- includes patients who continue to have signs and symptoms of heart failure despite optimal medical treatment.
- have advanced heart failure and are often evaluated for **heart transplant, ventricular assist devices, inotropic therapy**

Sign and symptoms

- ❖ Shortness of breath initially presents following physical exertion
- ❖ A chronic nonproductive cough is a nonspecific finding but is often present.
- ❖ Fatigue
- ❖ Leg edema and abdominal distension
- ❖ Loss of appetite caused by intestinal edema
- ❖ Diarrhea, weight loss, and visual changes

Stages of Heart Failure...

NEHYA- Classification

Functional class

- I Patients with cardiac disease but without limitations of physical activity. Ordinary physical activity does not cause undue fatigue, dyspnea, or palpitation
- II Patients with cardiac disease that results in slight limitations of physical activity. Ordinary physical activity results in fatigue, palpitation, dyspnea, or angina
- III Patients with cardiac disease that results in marked limitation of physical activity. Although patients are comfortable at rest, less than ordinary activity will lead to symptoms
- IV Patients with cardiac disease that results in an inability to carry on physical activity without discomfort. Symptoms of congestive heart failure are present even at rest. With any physical activity, increased discomfort is experienced

Compensatory Mechanisms in Response to Low Cardiac Output

- ❖ **Cardiac Contractility** results in a variety of compensatory mechanisms to achieve the metabolic demands of the body and bring it back toward normal function.
- ❖ **Sympathetic Nervous System Activation**
 - Activation of the sympathetic nervous system (SNS) and inhibition of the parasympathetic nervous
 - augmentation of cardiac contractility
 - heart rate
 - increase in the ventricular **preload** and **afterload**

Compensatory Mechanisms in Response to Low Cardiac Output...

❖ Renin-Angiotensin-Aldosterone System (RAAS)

Decrease in renal blood pressure

Causes the breakdown of angiotensinogen to angiotensin I

Angiotensin I is converted to Angiotensin II by the angiotensin-converting enzyme (ACE) located in the lungs

causing renal efferent vasoconstriction

- ❖ Prolonged angiotensin II release, however, results in peripheral vasoconstriction, which in turn results in increased **afterload**
- ❖ Consequently, the heart has to work harder and ultimately this results in worsening of **heart failure**.

Compensatory Mechanisms in Response to Low Cardiac Output..

Arginine Vasopressin

- ❖ Arginine vasopressin is responsible for regulating the plasma osmolality and volume.
- ❖ Vasopressin affects 2 types of receptors that are important from a cardiovascular actions:
- ❖ vasopressin type 1A (V1A) receptor and vasopressin type 2 (V₂) receptors.
 - V1A receptors are located on the cardiac myocytes and vascular smooth muscles. Stimulation of these receptors results in **vasoconstriction** and **inotropic effects**.
 - V₂ receptors are located in the **renal collecting ducts** and their stimulation results in **free water reabsorption**.
- ❖ Excessive vasopressin secretion in heart failure results in an increase in water retention thus contributing to vascular congestion that is a hallmark of congestive heart failure (CHF).

Compensatory Mechanisms in Response to Low Cardiac Output....

Left Ventricle Remodeling

- ❖ Loss of myocytes due to myocardial infarction
- ❖ Ventricular remodeling characterized by changes in the LV shape and size of the cavity
- ❖ Formation of scar tissue in the area of the infarction.
- ❖ progressive ventricular dilation
- ❖ decrease in contractile function

Pathophysiology of Heart Failure

Systolic dysfunction

- Occurs due to reduced cardiac output and significantly reduced ejection fraction ($EF < 45\%$; normal $> 60\%$), is typical of acute failure, especially that resulting from myocardial infarction.

$$EF = (EDV - ESV) / EDV * 100$$

where EDV=End diastolic volume and ESV is End systolic volume

Diastolic dysfunction

- often occurs as a result of **hypertrophy** and stiffening of the **myocardium**, and although cardiac output is **reduced**, EF may be normal. HF due to **diastolic dysfunction** does not usually respond optimally to positive inotropic drugs

Basic Pharmacology of Drugs Used In Heart Failure

Positive inotropes

- Drugs. Digitalis, B-agonists, Bipyridines

Digitalis

- Digitalis is the name of the genus of plants that provide most of the medically useful **cardiac glycosides**, eg, [digoxin](#)

- **Pharmacokinetics**

Absorption----65–80% absorbed after oral administration

Absorption of other glycosides varies from **zero** to nearly **100%**.

Distribution-----distributed to tissues, including the central nervous system (CNS).

Metabolism-----almost **two thirds** is excreted unchanged by the kidneys

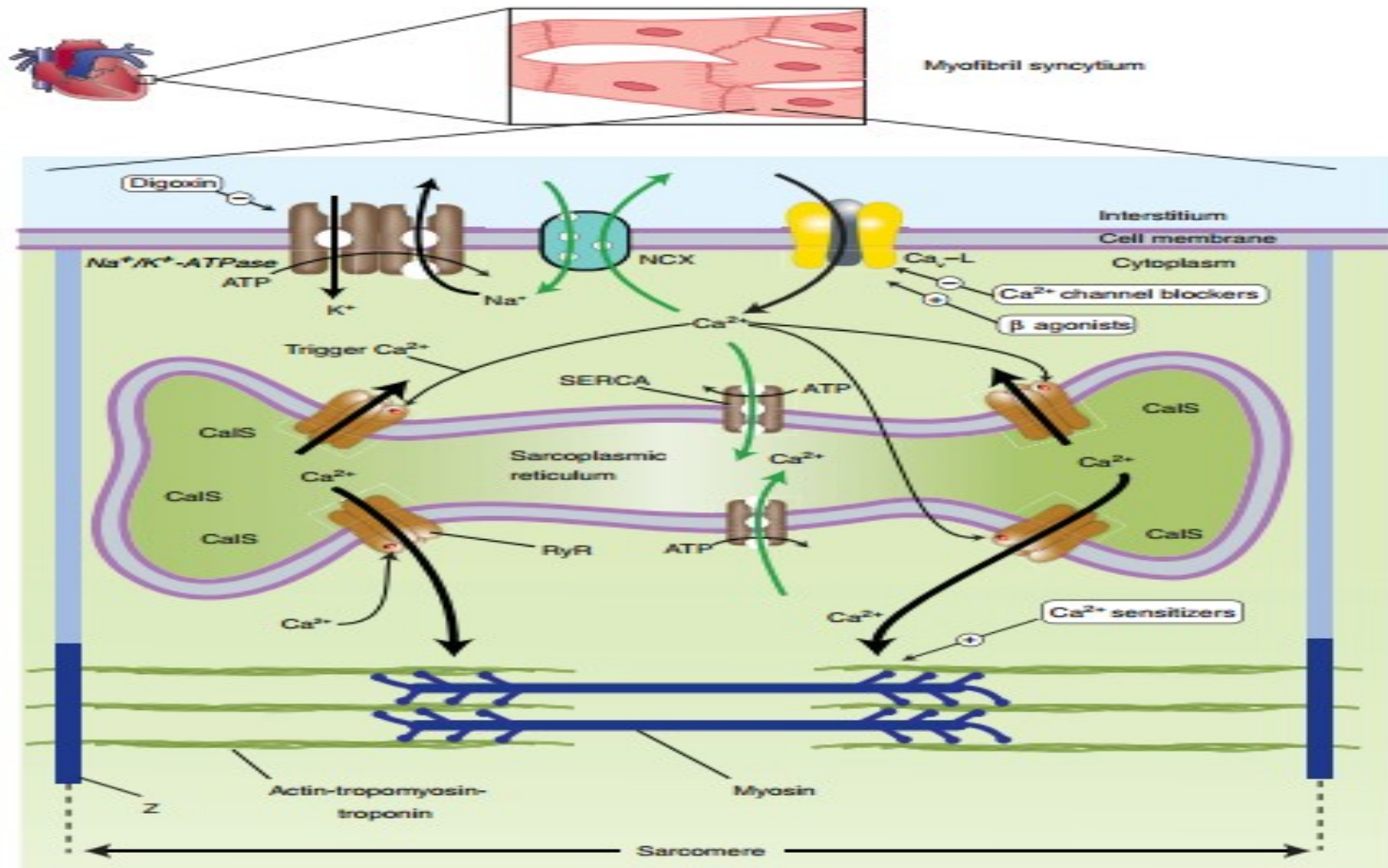
half-life is 36–40 hours

Pharmacodynamics

MOA

- Cardiac glycosides increase contraction of the cardiac sarcomere by increasing the free calcium concentration in the vicinity of the contractile proteins during systole.
- The increase in calcium concentration is the result of a two-step process:
 - **An increase of intracellular sodium** concentration because of Na^+/K^+ -ATPase inhibition; and second,
 - a relative **reduction of calcium expulsion** from the cell by the sodium-calcium exchanger NCX is caused by the increase in intracellular sodium

Pharmacodynamics....



Dose

Mild symptoms

- Slow loading (digitalization) with 0.125–0.25 mg/d or
- Rapid method (0.5–0.75 mg every 8 hours for three doses, followed by 0.125–0.25 mg/d).

Adverse drug reaction....The most common cardiac manifestations

- Atrioventricular junctional rhythm,
- premature ventricular depolarizations,,
- Ventricular tachycardia, and
- SA- blockade.

Adverse drug reaction...

Non- cardiac effects

- GI- effects....anorexia, nausea, vomiting, and diarrhea.
- CNS effects include vagal and chemoreceptor trigger zone stimulation.
- Less often, disorientation and hallucinations—especially in the elderly—and visual disturbances such as color perception.

Other positive inotropic

Bipyridines (Milrinone, inamrinone)

Milrinone

MOA---- inhibits phosphodiesterase isozyme 3 (PDE-3)....increase **cAMP** and has **vasodilation effect (both vein and artery)**

- It is active orally as well as parenterally but is available only in **parenteral** form.
- half-life of 3–6 hours, with 10–40% being excreted in the urine.

Pharmacodynamics

- increasing inward calcium flux in the heart during the action potential
- **Toxicity**.... nausea and vomiting; arrhythmias, thrombocytopenia, and liver enzyme changes (common with **inamrinone**)

Beta-adrenoceptor Agonists

Drugs... Dobutamine, Dopamine

➤ Dobutamine..

- produces an increase in **cardiac output**
- Some tachycardia and an increase in **myocardial oxygen consumption** have been reported.
- Therefore, the potential for producing **angina** or **arrhythmias** in patients **with coronary artery disease** is significant.

➤ Dopamine

has also been used in acute **heart failure** and may be particularly helpful if there is a need to **raise blood pressure**

Drugs without positive inotropic...

- These agents——are the first-line therapies for chronic heart failure.
- Drugs , ACE inhibitors, angiotensin receptor antagonists, aldosterone antagonists, and β blockers
- **Diuretics...furosemide is most commonly used**
 - They reduce salt and water retention, edema, and **symptoms**
 - **Major action in heart failure..... Reduce venous pressure and ventricular preload**
 - reduction of **cardiac size**, which leads to improved pump efficiency, is of major **importance in systolic failure**
 - **Spironolactone and eplerenone (aldosterone antagonists)...**have the additional benefit of decreasing morbidity and mortality in patients with severe heart failure

Angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, & related agents

- ACE inhibitors reduce peripheral resistance and thereby reduce **afterload**;
- they also reduce salt and water retention (by reducing aldosterone secretion) and in that way reduce **preload**.
- The reduction in tissue **angiotensin levels** also reduces sympathetic activity.

Angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, & related agents...

- Angiotensin receptor blockers such as **losartan** to have similar beneficial effects.
- Angiotensin receptor blockers should be considered in patients intolerant of ACE inhibitors because of **incessant cough**
- **Aliskiren**, a renin inhibitor approved for hypertension, was found to have no definitive benefit in clinical **trials for heart failure**.

Vasodilators

- Vasodilators are effective in acute heart failure Through
 - a reduction in **preload** (through venodilation), or
 - reduction in **afterload** (through arteriolar dilation), or both.
- long-term vasodilation by hydralazine **and isosorbide dinitrate** can also reduce damaging remodeling of the heart.
- Vasodilator drugs can be divided into selective **arteriolar dilators, venous dilators**, and drugs with nonselective vasodilating effects

Vasodilators....

- Drug selection is depend up on sign and symptoms
- patients with high filling pressures in whom the principal symptom is dyspnea, venous dilators such as long-acting **nitrates are used.**
- For patients with fatigue due to low left ventricular output an arteriolar dilator may be helpful in increasing **forward cardiac output**

Beta-adrenoceptor Blockers

- Most patients with chronic heart failure respond favorably to certain β blockers
- But these drugs can precipitate acute **decompensation** of cardiac function
- suggested mechanisms
 - attenuation of the **adverse effects** of high concentrations of catecholamines (including apoptosis),
 - up-regulation of β receptors
 - decreased heart rate
 - reduced remodeling through inhibition of the mitogenic activity of catecholamines.